



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Business systems

Academic Year 2024-2025 (Even Semester)

Degree, Semester & Branch: VI Semester B.Tech -CSBS

Course Code & Title: CCS339-Cryptocurrency and Blockchain Technologies

Name of the Faculty member (s): Mrs. B.Yazhini

Innovative Practice Description

Unit / Topic: Unit I/ INTRODUCTION TO BLOCKCHAIN

Course Outcome: CO 1

Learning Outcome: 1a,1b,1c

Activity Chosen: Mind Map

Justification:

- A mind map is a visual representation that shows concepts and ideas. It's a technique for visual thinking that helps with data arrangement.
- It swiftly jots down ideas. So, this practice will help students recognize the theoretical ideas in the image and facilitate quick recall for their tests.
- Time Allotted for the Activity: 15 minutes

Details of the Implementation:

- Instructor explained the particular concepts/topic in classroom within 35minutes.
- Based on the discussion and the teacher asked the students to draw a mind map related to the topic within 15 minutes.
- Each student created a mind map on Dictionaries: operation and methods.
- Students represent the concept taught in the class by visual representation.
- Instructor collected the sheets from the students. After that they are asked to discussed about their mind map with their neighbours.

CO — PO / PSO mapping:

CO	POI	P02	P03	P04	P05	P03	PSOI	PSO3
CO 1	3	3	2	2	1	1	3	1

(1 — Low 2 — Moderate 3 — High)

PO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO4	PO5
	2	1	1	1	1	1
Justification for correlation	Students will acquire in-depth knowledge of the abstract models of blockchain technology by applying engineering fundamentals	Students will analyze blockchain use cases and scenarios using engineering sciences	Students will design and develop blockchain based solutions by implementing smart contracts and decentralized applications	Students will investigate complex problems related to abstract blockchain models, exploring potential improvements and innovative approaches.	Students will investigate complex problems related to abstract blockchain models, exploring potential improvements and innovative approaches.	Students will effectively use blockchain development tools and platforms to implement and gain hands-on experience in blockchain technology.

PSO mapped:

Innovative practice	PSO1	PSO3
	3	1
Justification for correlation	To apply analytic technologies to arrive at actionable foresight, Insight, hindsight from data for solving business and engineering problems	To enrich the critical thinking skills in emerging technologies such as Hybrid Mobile application development, cloud technology stack, and cyber-physical systems with mathematical aid to foresee the research findings and provide the solutions

- Images / Screenshot of the practice:

BLOCK CHAIN MINDMAP

BLOCK PROPAGATION:

Block propagation refers to the process by which a newly mined block is transmitted across the blockchain network to all other nodes.

TECHNIQUES:

COMPACT BLOCKS: Share only the block header and short transaction IDs.

GRAPHENE: Uses a more compressed method to minimize data transmission.

HOW IT WORKS:

A miner finds a valid block.

The block is immediately sent to neighboring nodes.

The process continues until the block is known to the entire network.

GOALS

SPEED: Faster propagation reduces the chances of orphaned blocks.

EFFICIENCY: Bandwidth and resource usage should be optimized.

BLOCK RELAY:

Block relay is a method of forwarding block between nodes, typically in a optimized or structured way.

TYPES OF BLOCK RELAY

STANDARD RELAY: Peer to Peer transmission using gossip protocol.

FAST RELAY NETWORK: These are high-speed, often centralized or semi-centralized relay networks ensure faster.

- Sample mindmap:

Fig:1 Glimpse of Student Mind Map

- Reflective Critique:

Feedback of practice from students and other stakeholders:

- Students expressed that they understood the concept very well and the pictorial representation made them to understand the topic clearly.
- This helped the students in recalling the topic taught on that specific day, generating new ideas about the topic.

Benefit of the practice:

- From this activity, the students can get more clarity in the particular topic.
- Through this activity, the students are able to remember and describe the various dictionary operations and its methods clearly. Challenges faced in implementation:
- Some students find it challenging to illustrate the ideas with pictures.
- Several of them require additional explanations of dictionary operations. • Several examples are provided so they can easily understand the subject.

References:

<https://www.ritrjpm.ac.in/images/computer-science/Mind%20Map.pdf>

<https://www.lucidchart.com/pages/how-to-make-a-mind-map>

Faculty Incharge

HOD

